

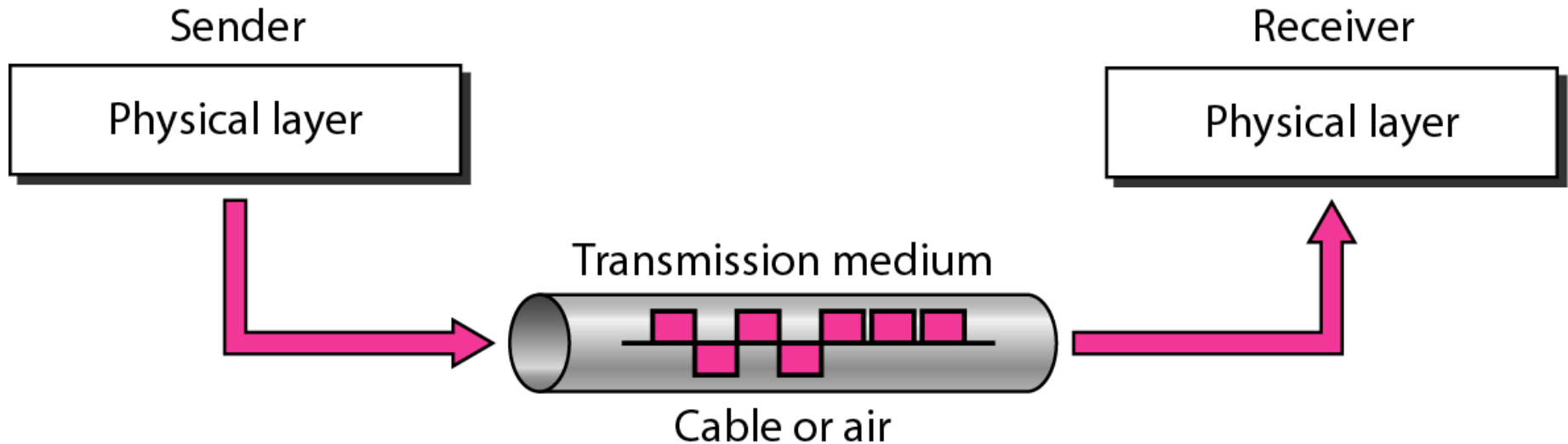


Computer Network

Chapter 4 : Transmission Media

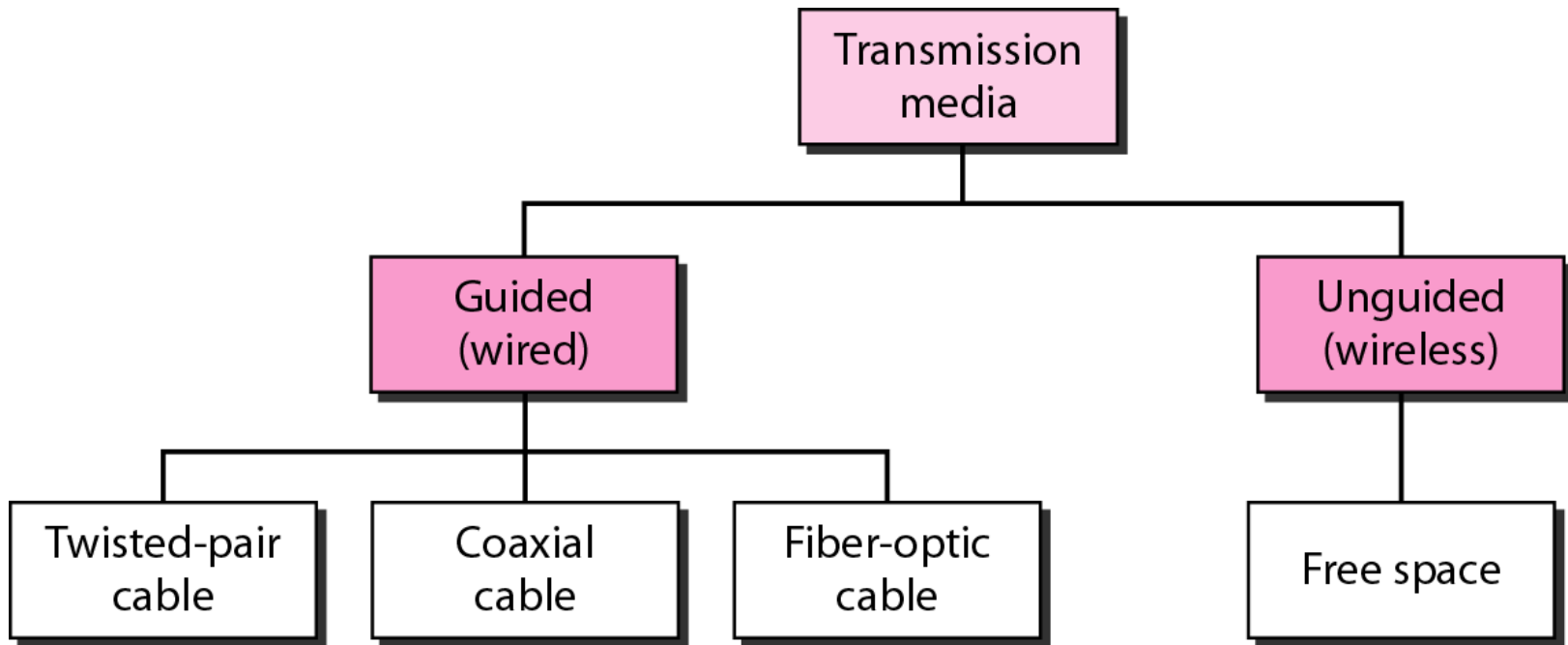


Transmission medium and physical layer





Classes of transmission media





GUIDED MEDIA

Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

Topics discussed in this section:

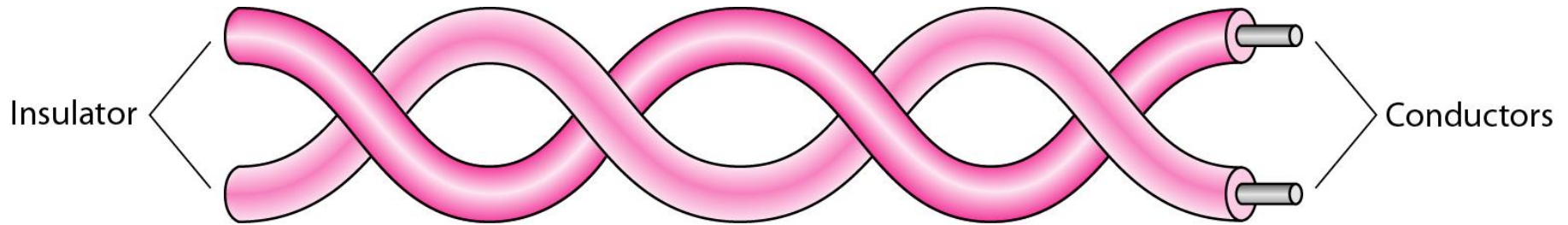
Twisted-Pair Cable

Coaxial Cable

Fiber-Optic Cable



Twisted-pair cable

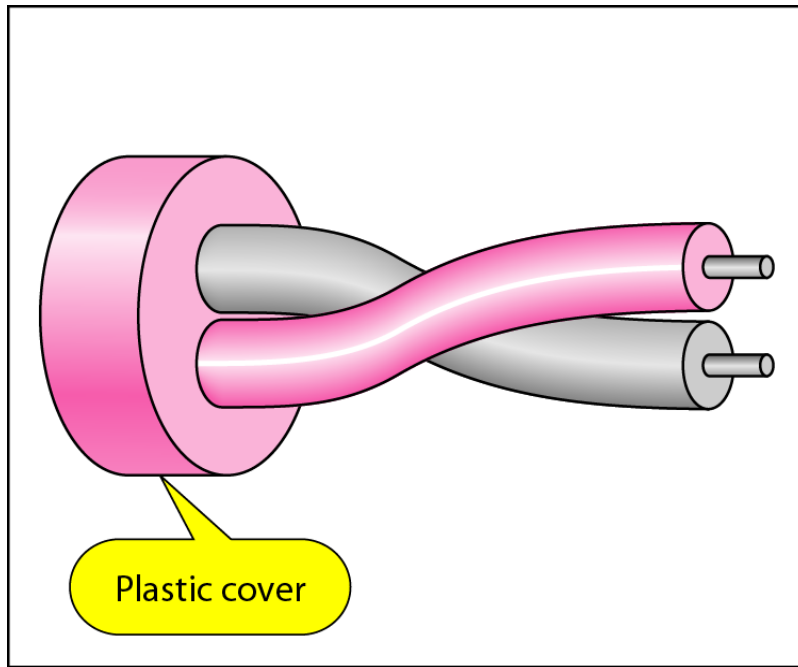


Why twisted?

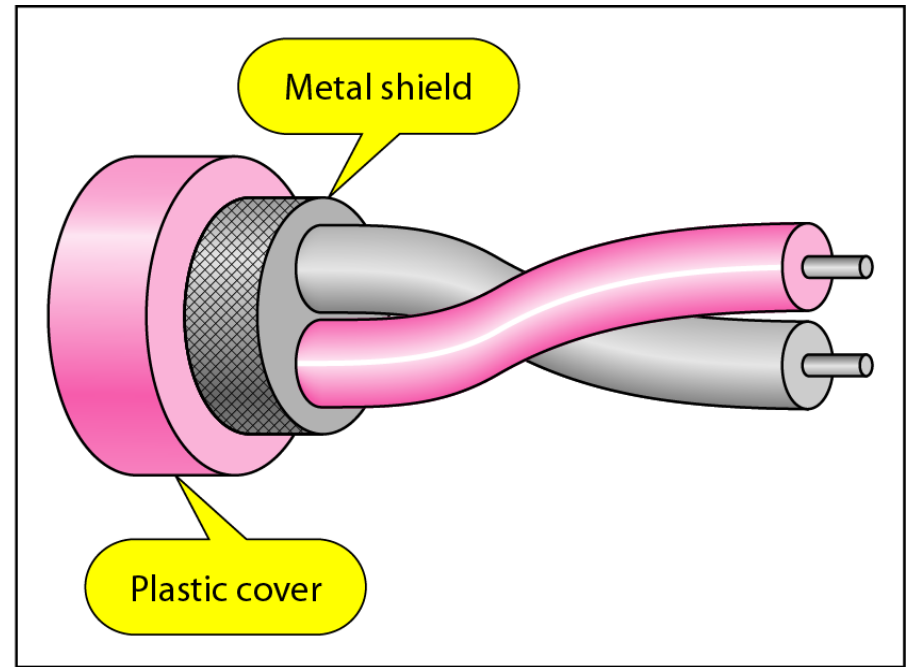
To make unwanted signals interference cancel out each other.



UTP (unshielded twisted pair) and STP (shielded twisted pair) cables



a. UTP

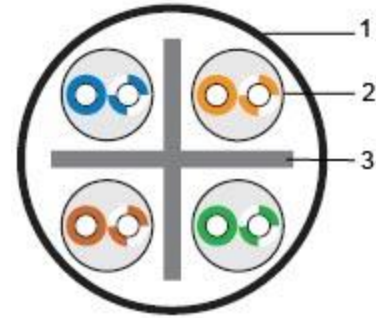
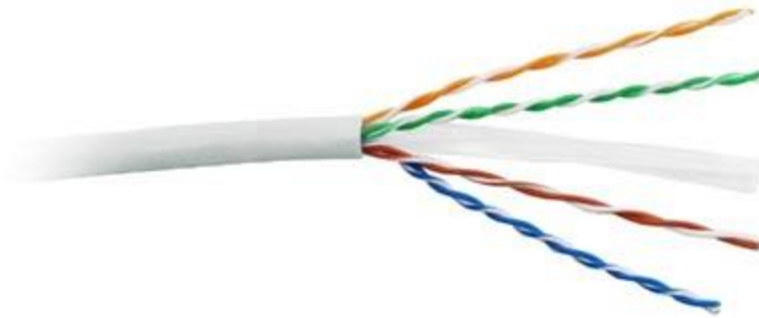


b. STP

Example: Ethernet, phone line



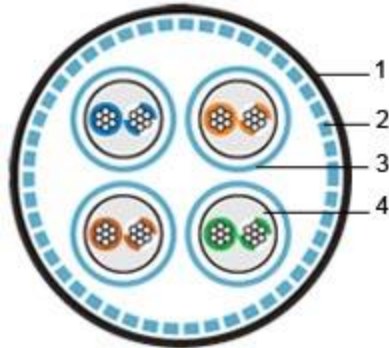
UTP cables



- 1 – Jacket
- 2 – Solid twisted pair
- 3 – Spacer



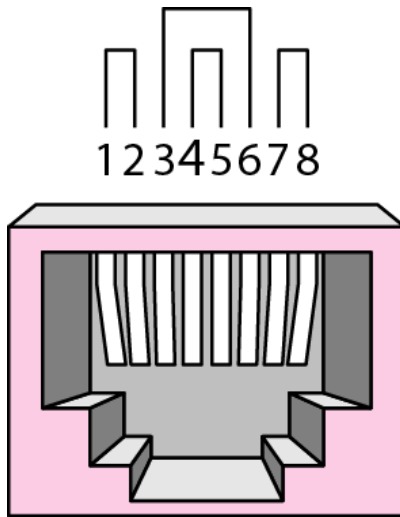
SSTP cables



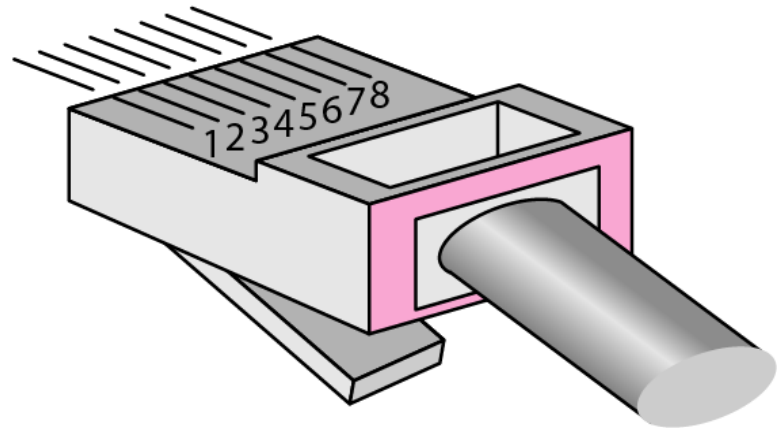
- 1 – Jacket
- 2 – Shield-braid
- 3 – Shield-foil
- 4 – Stranded twisted pair



UTP connector



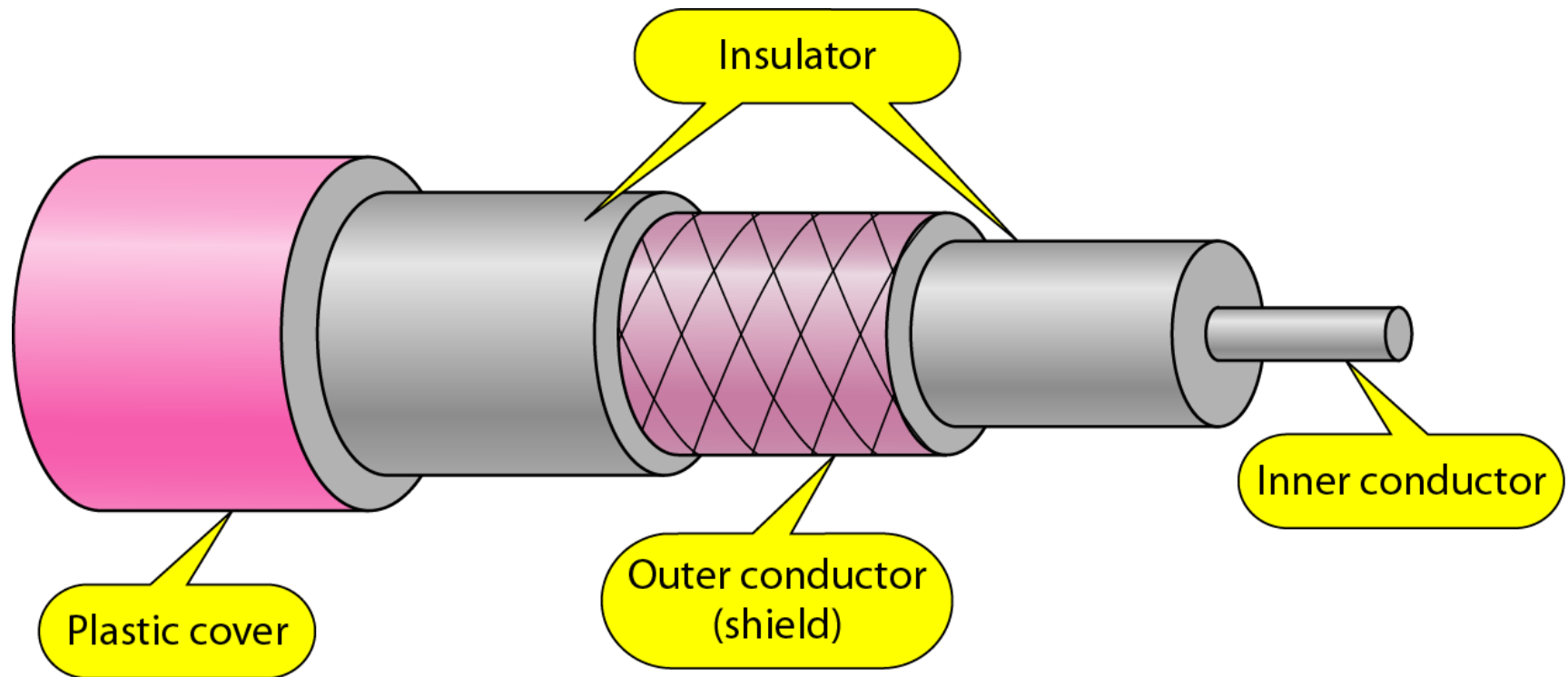
RJ-45 Female



RJ-45 Male

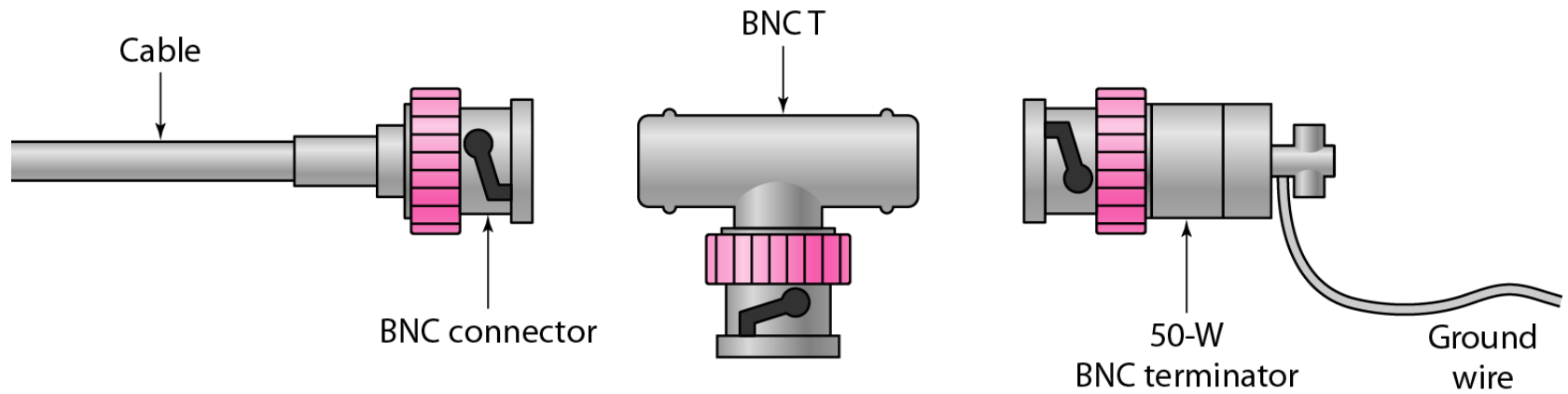


Coaxial cable





BNC connectors





Twisted-pair cable vs. coaxial cable



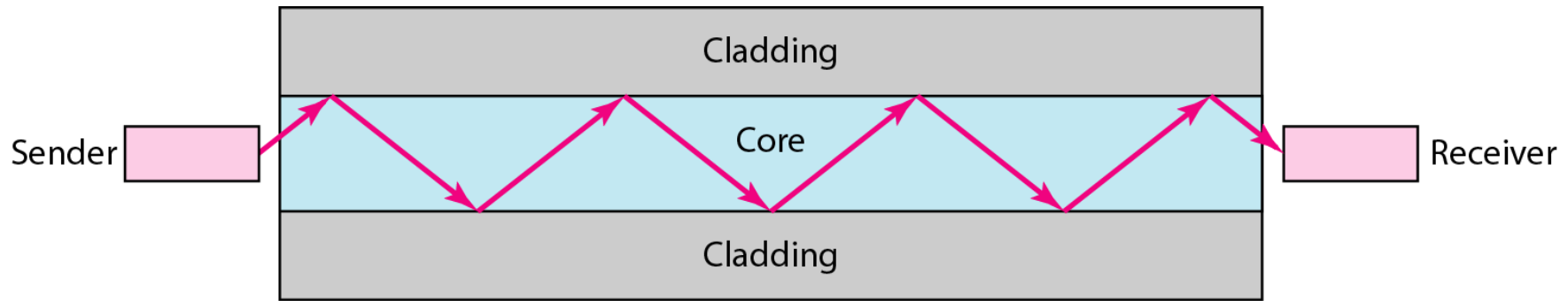
Bandwidth: coaxial > twisted-pair

Transmission distance: twisted-pair > coaxial

Thus cable needs frequent use of repeaters.



Optical fiber

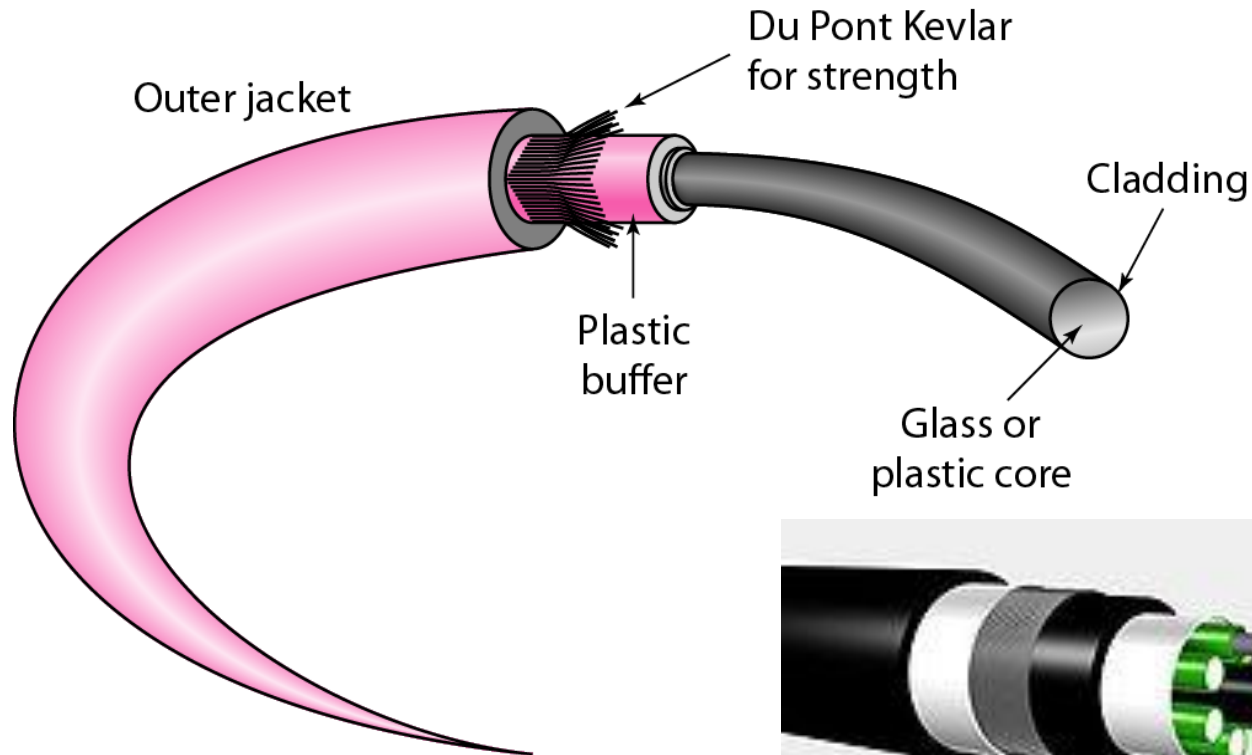


This is the reason why optical fiber cannot be bended arbitrarily.



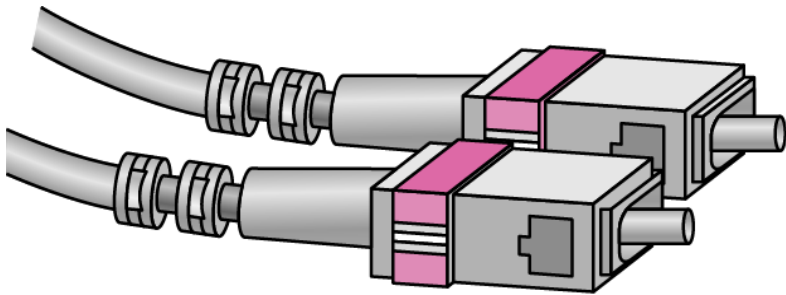


Fiber construction

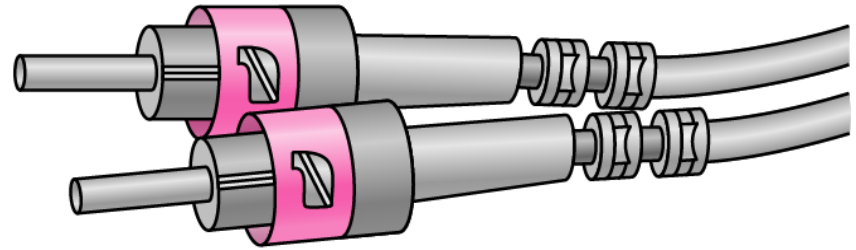




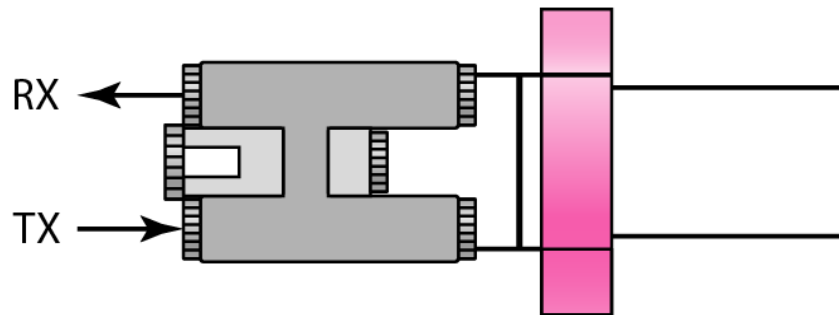
Fiber-optic cable connectors



SC connector



ST connector



MT-RJ connector





Pros and Cons for Optical Fiber Cable

Pros:

- *Higher bandwidth*
- *Less signal attenuation (50km without repeater; twisted pair and coaxial requires 5km per repeater)*
- *Immune to electromagnetic interference*
- *Resistance to corrosive materials*
- *Light weight*
- *Good resist to tapping*

Cons:

- *Installation and maintance*
- *One direction communication for one line (not duplex)*
- *Cost*



LAN

- Connect devices in a small area
- Segment length

Table 2-1 *Ethernet Segment Distance Limitations*

Ethernet Specification	Description	Segment Length
10BASE-T	10-Mbps Ethernet over twisted-pair	100 m
10BASE-FL	10-Mbps over fiber-optic cable	2000 m
100BASE-TX	100-Mbps Ethernet over twisted-pair	100 m
100BASE-FX	Fast Ethernet, still 100-Mbps, over fiber-optic cable	400 m
1000BASE-T	Gigabit Ethernet, 1000-Mbps, over twisted-pair	100 m
1000BASE-LX	Gigabit Ethernet over fiber-optic cable	550 m if 62.5-micron (μ) or 50- μ multimode fiber; 10 km if 10- μ single-mode fiber
1000BASE-SX	Gigabit Ethernet over fiber-optic cable	250 m if 62.5- μ multimode fiber; 550 m if 50- μ multimode fiber
1000BASE-CX	Gigabit Ethernet over copper cabling	25 m

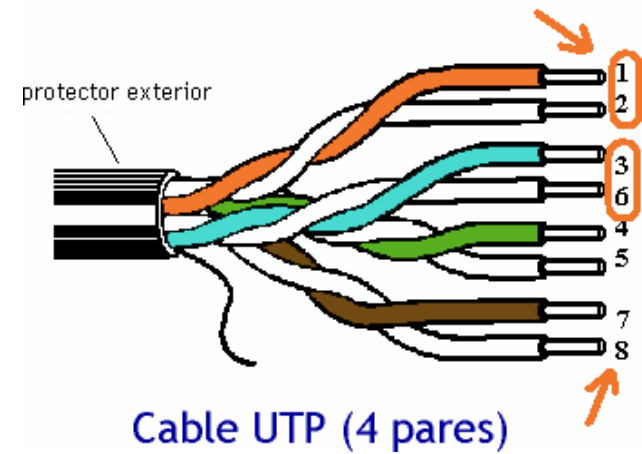
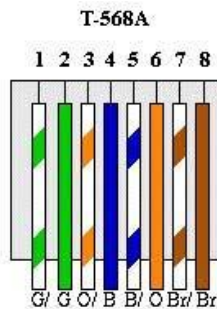
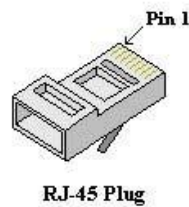
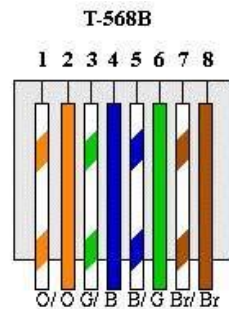


Cable Installation





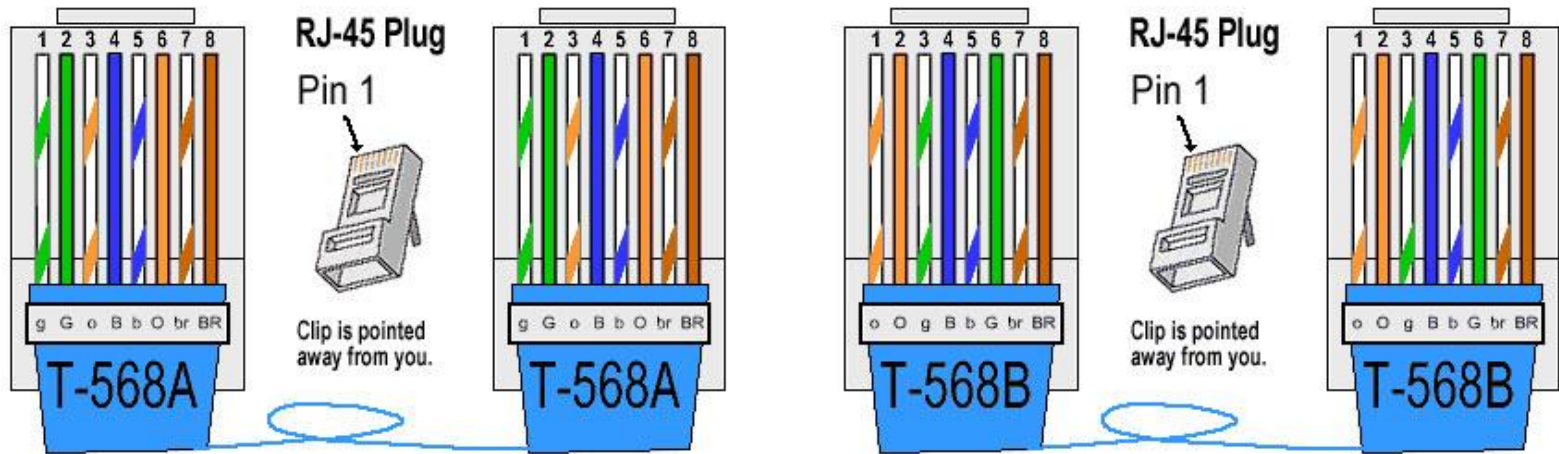
Cable Installation



- T568A and T568B



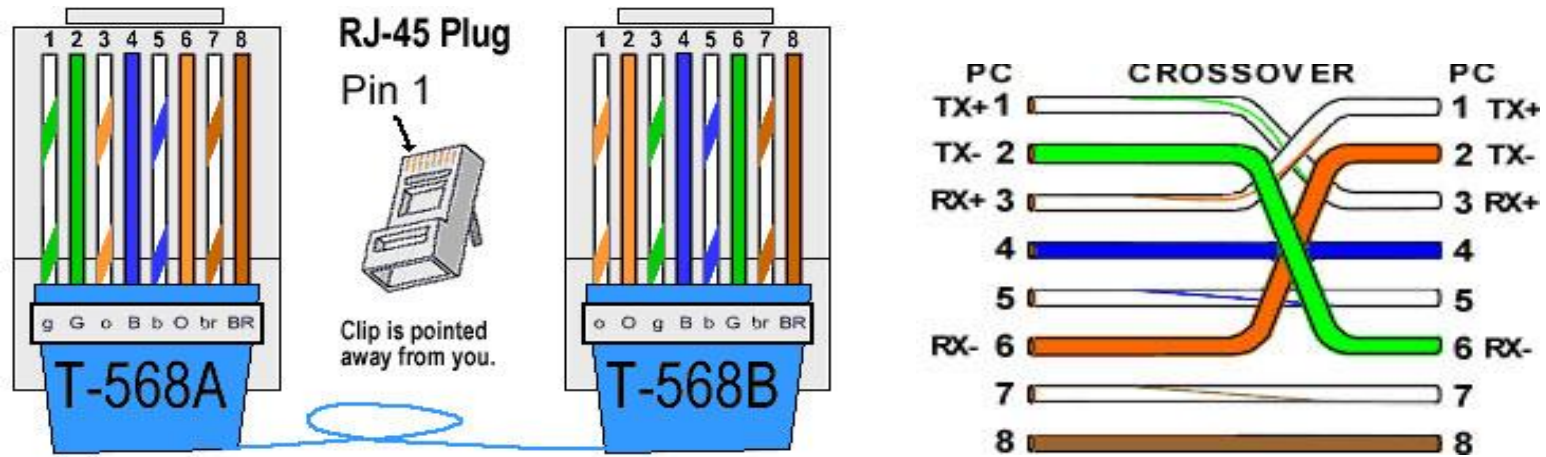
Cable Installation



- Straight-Through cable: PC vs Hub, Switch (10/100 Base-T)



Cable Installation



- Crossover cable: PC-PC, Hub - Hub, Switch - Switch.



UNGUIDED MEDIA: WIRELESS

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.

Topics discussed in this section:

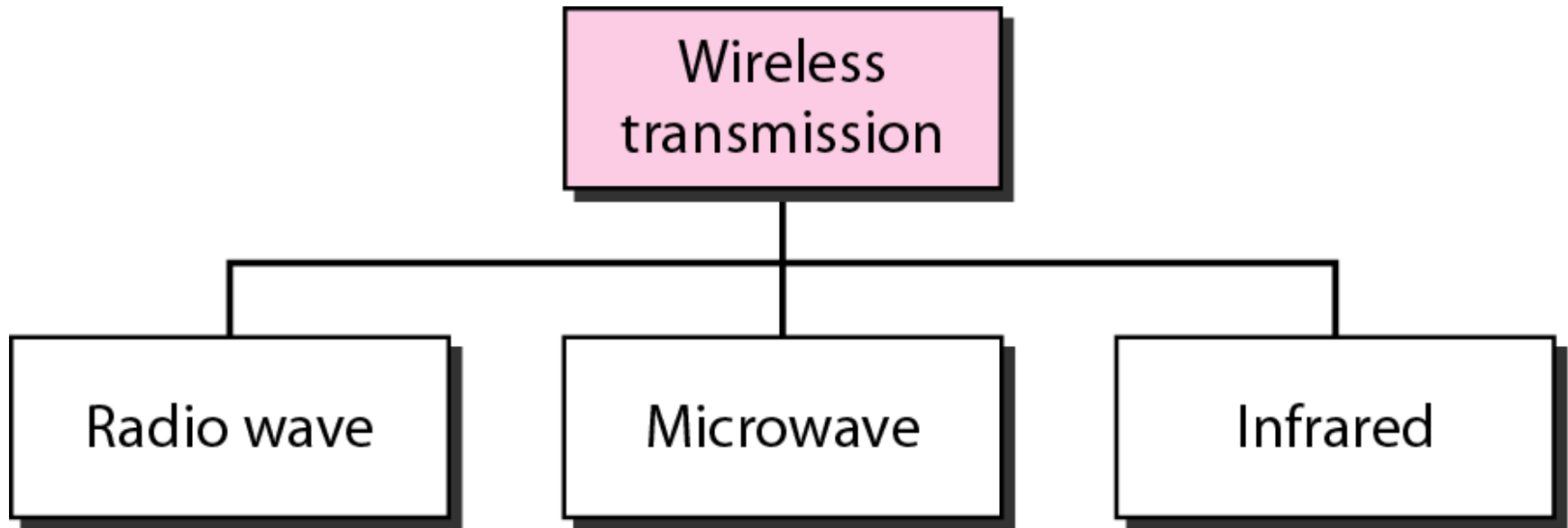
Radio Waves

Microwaves

Infrared

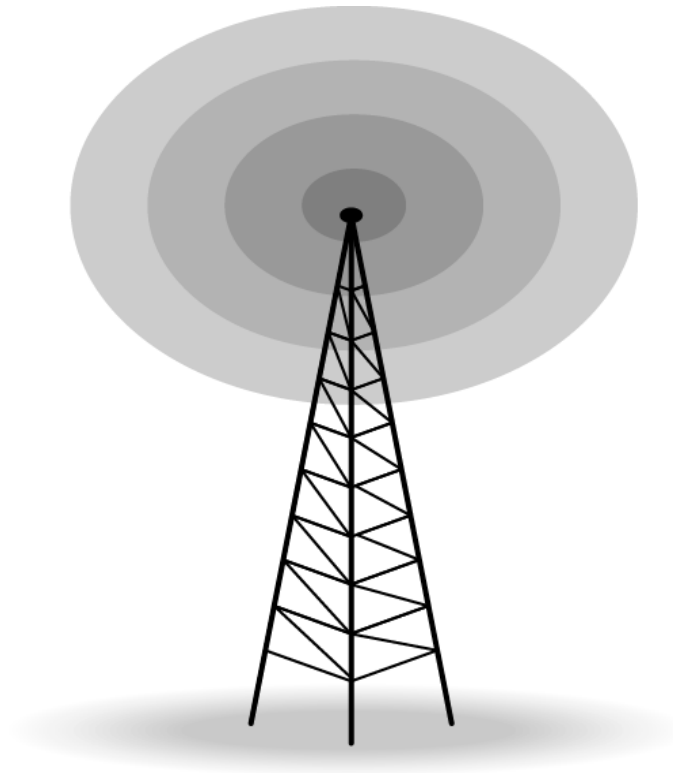


Wireless transmission waves





Omnidirectional antenna





Example: Omnidirectional antenna



Use:
Wireless router



Advantages and Disadvantages Wireless Communication

- Advantages
 - User Mobility
 - Easy to install
 - Reduced cost
 - Scalability
- Disadvantages
 - High data error rate
 - Lower transmission data rates
 - Security
 - Battery of Mobile Devices
 - Health Issues